

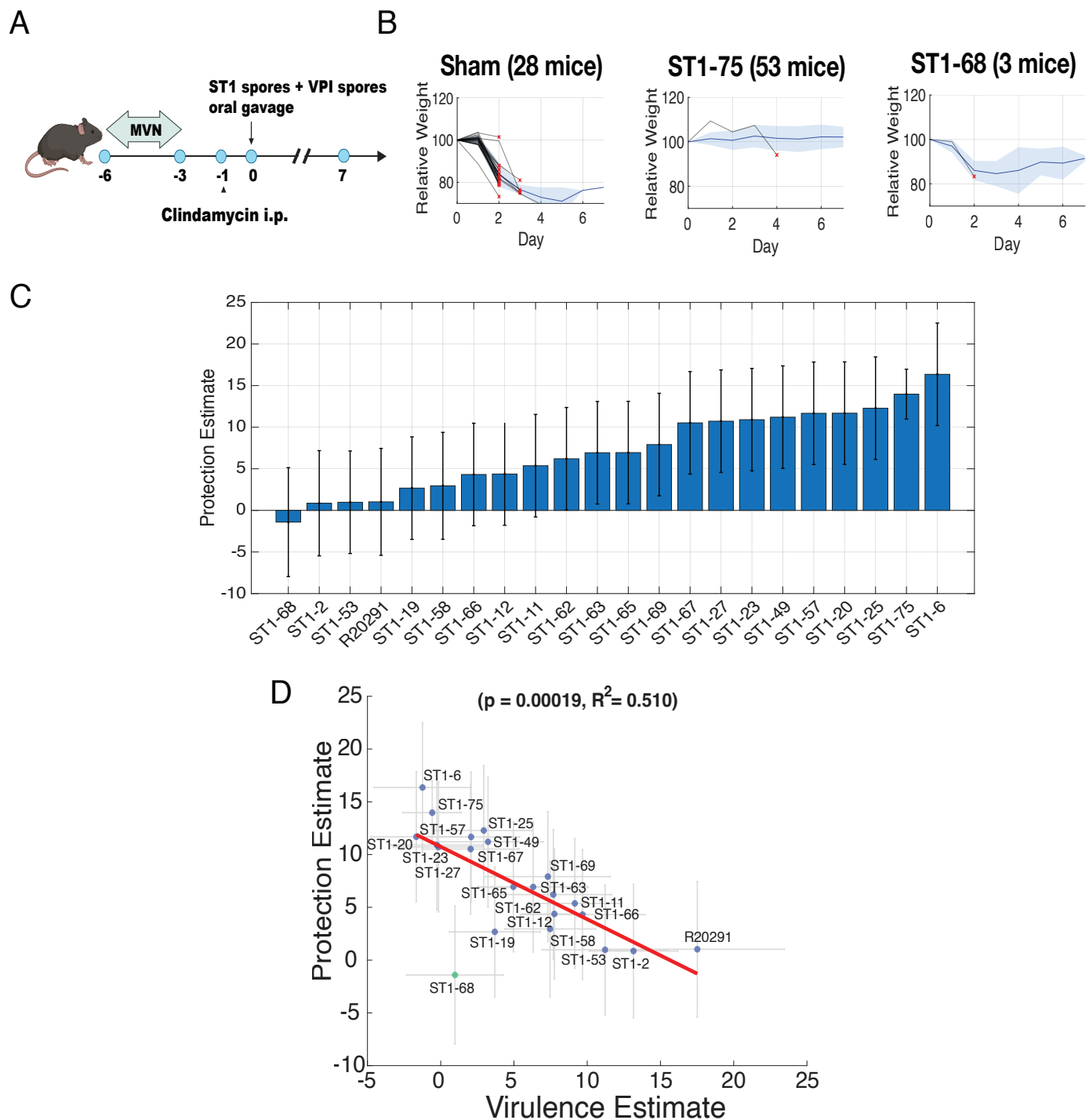


Figure 1: (A) Schematic of the co-infection mouse model used to assess protection. Mice were given the cocktail of antibiotics, metronidazole, neomycin and vancomycin for 3 days, followed by antibiotic cessation for 2 days. Mice were then given clindamycin intraperitoneally then co-infected with VPI10463 spores together with spores of an individual ST1 strain via oral gavage a day later. (B) Representative relative weight trajectories during co-infection for sham+VPI (PBS; n=28), ST1.75+VPI (n=53), and ST1.68+VPI (n=3) groups. Shaded regions indicate 95% confidence intervals. (C) Protection scores for all 21 ST1 clinical isolates, derived from linear mixed-effects models of relative weight loss during VPI10463 co-infection. The model used sham+VPI mice as the reference, such that positive effect sizes reflect weight preservation (greater protection). Error bars indicate standard error. (D) Relationship between virulence score (x-axis) versus protection score (y-axis) for all 21 ST1 isolates. A significant negative linear relationship was observed ($R^2 = 0.510$, $p = 0.00019$) shown here in red line, indicating that higher virulence correlates with reduced protection against VPI10463. The outlier ST1.68 strain, colored in green, disagreed with that trend (statistically significant outlier by Bonferroni-corrected externally studentized residual test ($t = -5.16$, $p = 0.0012$))